Flame Transfer Functions for Liquid-Fueled Swirl-Stabilized Turbulent Lean Direct Fuel Injection Combustion

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Heat release rate responses to inlet fuel modulations, i.e., the flame transfer function (FTF), are measured for a turbulent, liquid-fueled, swirl-stabilized lean direct fuel injection combustor. Fuel modulations are achieved using a motor-driven rotary fuel valve designed specially for this purpose, which is capable of fuel modulations of up to 1 kHz. Small-amplitude fuel modulations, typically below 2.0% of the mean fuel, are applied in this study. There is almost no change in FTFs at different fuel-modulation amplitudes, implying that the derived FTFs are linear and that the induced heat release rate oscillations mainly respond to variations in the instantaneous fuel flow rate rather than in the droplet size and distribution. The gain and phases of the FTFs at different air flow rates and preheat temperatures are examined. The instantaneous fuel flow rate is determined from pressure measurements upstream of a fuel nozzle. Applications of the FTF to modeling and control of combustion instability and lean blowout are discussed.
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1 Introduction

Combustion instability and lean blowout are major technical challenges for both gas-fueled and liquid-fueled dry-low-emission (DLE) combustion [1–3]. Combustion instability refers to the self-excited, large-amplitude, limit-cycle thermoacoustic oscillations caused by positive feedback between pressure and heat release rate. Strong pressure pulsations shorten an engine's life due to enhanced heat transfer, lower combustion efficiency, exacerbate NO_x emissions, generate noise pollutions, and may even cause instrumentation failures. Lean blowout (LBO) refers to partial or global flame quenching when the equivalence ratio is decreased below certain thresholds. As the equivalence ratio approaches the lean flammability limits, a combustor's resistance to external disturbances or small deviations from the equilibrium points is considerably weakened [4]. Thus, a small external disturbance in the inlet turbulence, the fuel composition, the fuel flow rate, and/or the air flow rate may quench the flame. LBO necessitates expensive shutdown and restarting procedures for land-based gas turbines, and represents a major safety hazard for aircraft engines.

The underlying mechanisms for combustion instability and LBO are fundamentally different, but they both make engines unable to operate in an efficient, safe, and environment-friendly manner at lean conditions. Model-based control design is a standard approach for control engineers and theorists. First-principle low-order models provide valuable insight into the underlying mechanisms, but their accuracy critically depends on the assumptions and simplifications [5]. The measured flame transfer functions (FTFs), i.e., heat release responses to inlet air and/or fuel modulations, provide an accurate description of combustion dynamics around the working conditions where they are derived.

The FTF is determined by both the intrinsic dynamics of a specific combustor and the manner in which dynamic fuel is introduced into and dispersed inside the reacting flow field. To minimize the amount of fuel modulations required, it is desirable for the FTF to have as large a gain as possible. This can only be achieved by maximizing the conversion of modulated fuel flow rate to modulated heat release rate. In gas turbine combustors, the heat release zone lies along the swirling shear layer outside the recirculating vortex breakdown. Because of flow field inhomogeneities and the convective time delay, the spatially and temporally coherent fuel modulations exiting the fuel nozzle become spatially dispersed and temporally less coherent as they are transported downstream, which lowers the FTF gain. The situation becomes even worse for longer flames. Droplet size also plays a role in the resulting FTFs. Yu and Wilson [6] noticed that the critical fuel flux, i.e., the minimum fuel modulations required to attenuate combustion instability, depends on the droplet size. Smaller droplet size reduces the heating and evaporation time, and enhances the control effectiveness. This observation is consistent with the proposition that the dynamic evaporation process may play an important role in liquid-fueled combustion instability [7]. Droplet size and distribution can be measured using phase Doppler particle analyzer (PDPA), Malvern sizers, combined Mie-scattering/planar laser-induced fluorescence (PLIF), and other means. However, this is out of the scope of the present study.

This paper is organized as follows. First, the experimental setup, including the combustion rig, optical setup, and the high-frequency rotary valve, is described. Second, a method for determining the instantaneous fuel flow rate out of a fuel nozzle is presented. This method is based on acoustic wave theory and pressure measurements upstream of the fuel nozzle. Third, the FTFs at different air flow rates and preheat temperatures are reported. At last, we discuss potential applications of FTFs to combustion analysis and control, including combustion instability and lean blowout.

2 Experiment Setup

Figure 1 shows the combustion rig, optical setup, and the rotary fuel valve. Preheated air enters a quartz combustion chamber, which is 0.10 m in diameter and 0.30 m in length, through a 30 deg radial-entry swirler. The exit diameter of the swirler is 3.56 cm. Pressure drop across the air swirler varies with the air

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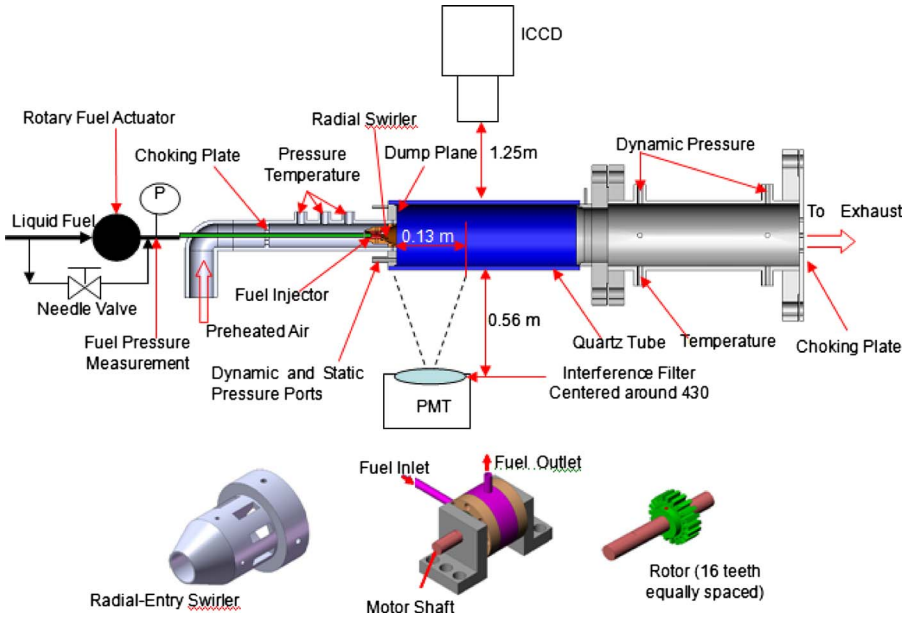

Fig. 1 The combustion rig, optical setup, and the rotary fuel actuator

flow rate and preheat temperature, but it is within 5% for all results reported here. 1-Decene or Jet-A is injected into the swirling air flow 0.02 m upstream of the dump plane using a single-point macrolaminated fuel nozzle (2.5 GPH) from Parker Hannifin. The dump plane refers to the solid surface between the swirler exit and the quartz combustion chamber. With a pressure drop of 689 kPa across the fuel nozzle, the droplet Saunter mean diameter (SMD) is less than 10 μm [8]. The air flow rate is measured using a vortex flowmeter from Omega, Stamford, CT, and the measurement uncertainty is within 1% of reading. A photomultiplier tube (PMT) (H7732-10 from Hamamatsu Photonics, Bridgewater, NJ) featured with an interference bandpass filter ($\sim$430 nm or $\sim$307 nm with full width half minimum (FWHM) of 10 nm) is used for global chemiluminescence measurements. Generally speaking, chemiluminescence should not be assumed proportional to the instantaneous heat release rate. Probably the proportionality only exists in the weakly turbulent or the wrinkled flamelet region and in the absence of equivalence ratio variations and strong acoustic oscillations [9]. However, the instantaneous chemiluminescence intensity reflects the dynamics of the air consumption rate, the equivalence ratio, the flame temperature, and acoustic motions, thus it can be used as a valuable indicator of heat release dynamics. The PMT is placed 0.06 m downstream of the dump plane and 0.56 m away from and perpendicular to the quartz chamber. The PMT viewing area is within 0.13 m downstream of the dump plane. Time scale analysis shows that the droplets evaporation time is smaller than the convection time. Thus before the droplets are transported to the flame front, they may have been fully evaporated. In addition, from Fig. 1 one can see that high-speed air enters the diverging swirler through eight radial-entry slots, mixes with fine droplets, recombines into one stream, and suddenly expands at the dump plane. Expectedly, droplet/air entrainment and fuel/air mixing are extremely fast, and a rather homogeneous mixture of fuel vapor and air may form upstream of the major heat release zone. Thus combustion in the present rig mainly occurs in the lean premixed mode [9].

Fuel modulations are achieved using a motor-driven rotary fuel valve designed specially for this purpose. The fuel-modulation frequency is determined by the number of rotor teeth and the motor rotating speed. Liquid fuel is stored in a 19-l stainless steel tank and pressurized with N_2 at 1378 kPa. The mean fuel flow rate is measured using a linear flow meter, and the measurement uncertainty is within 5% of reading. The mean fuel flow rate is controlled using a needle valve. The rotary valve is located 0.46 m upstream of the fuel nozzle. A 0.8-l accumulator is located 0.2 m upstream of the rotary valve. Two Sensotec pressure transducers (sensitivity 172 kPa/V and uncertainty 0.25% of reading) capable of working up to 2 kHz are installed 0.29 m and 0.45 m upstream of the fuel nozzle, respectively. A bypass flow passage is installed parallel to the rotary fuel valve, and fuel-modulation amplitude is controlled by varying fuel splits between the rotary fuel valve and the bypass passage. The fuel-modulation ratio, which is defined as the ratio of the root-mean-square (rms) to mean fuel line pressure, can be greater than 30% up to 750 Hz and greater than 15% up to 1 kHz.

3 Determination of the Instantaneous Fuel Flow Rate

The instantaneous fuel flow rate out of a fuel injector is an important parameter for combustion instability analysis and control. Unsteady fuel flow rates introduced into a combustor can be caused by both self-excited pressure pulsations and fuel modulations. Because of the harsh environments and limited space inside the plenum, pressure immediately upstream of the fuel nozzle cannot be measured. Pressure measurements at some points upstream of the fuel nozzle may differ considerably in both amplitude and phases from those immediately upstream of the fuel nozzle. A literature survey shows that accurate determination of the instantaneous fuel flow rate has not been well addressed. In this study, we determine fuel modulations based on acoustic wave theory and pressure measurements at two locations upstream of the fuel nozzle. This method is an extension of a method for determining fuel flow rate modulations out of a gaseous fuel nozzle [10]. Although the mathematical formulations are essentially the same for the two methods, two major differences exist. First, the speed of sound for liquid fuel is typically above 1000 m/s [11], which is much higher than that of gaseous fuel. In addition, in order to improve atomization quality, pressure drop across a liquid-fueled pressure-swirl nozzle is typically much higher than that across a gaseous fuel injector. Thus a liquid-fueled pressure-swirl can be accurately modeled as a purely resistive element, but a gaseous fuel nozzle may not be. In the case of small-amplitude fuel modulations, the flow pattern across the fuel

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nozzle can be assumed invariant, thus perturbations in the instantaneous fuel flow rate can be determined as

$$\frac{\dot{m}_f'}{\bar{\dot{m}}_f} = \frac{p_0(t)}{2\overline{\Delta P}} = \frac{\mathrm{Re}(\tilde{P}(0,t))}{2\overline{\Delta P}} \tag{1}$$

In the case of large-amplitude fuel modulations, in particular, at high frequency, the 1D nonlinear wave equation accounting for nonlinear acoustics and viscous shear stress should be used. Solutions to nonlinear wave equations are less straightforward than those to linear waves. A simple method for validating the linear acoustic assumption is to install a pressure transducer at another location in the fuel line and to compare the measured pressure with the linear prediction. It is found that, even without accounting for nonlinear acoustics and viscous shear stress, the developed method can well predict large-amplitude pressure oscillations of up to 20% below 300 Hz, and the prediction error is within 6% in amplitude and within 6 deg in phases [10].

4 Flame Transfer Functions

The FTF can be defined in multiple forms, such as,

$$W_m(s = i\omega) = \mathrm{CH}^*(s)/P_m(s) \tag{2}$$

$$W_0(s = i\omega) = \mathrm{CH}^*(s)/P_0(s) \tag{3}$$

where $\mathrm{CH}^*(s)$, $P_m(s)$, and $P_0(s)$ refer to the Laplace transform of the overall CH^* chemiluminescence intensity, pressure measurements 0.29 m upstream of the fuel nozzle, and fuel pressure immediately upstream of the fuel nozzle, respectively. The relationship between $P_m(s)$ and $P_0(s)$ can be determined from the acoustic wave equations as follows [11]:

$$W_N = \frac{P_m(s)}{P_0(s)} = \frac{\tilde{A}e^{-is(x_m/\bar{c})} + \tilde{B}e^{is(x_m/\bar{c})}}{\tilde{A} + \tilde{B}} = \frac{\tilde{A}/\tilde{B}e^{-is(x_m/\bar{c})} + e^{s(x_m/\bar{c})}}{\tilde{A}/\tilde{B} + 1} \tag{4}$$

At a fuel flow rate of 1.4 g/s, $\tilde{A}/\tilde{B}$ is determined as 1.80. The magnitude of W_N consistently decreases with frequency from unity at 0 Hz to 0.65 at 600 Hz, and the phase lead consistently increases with frequency from zero at zero frequency to 20.5 deg at 600 Hz. During fuel modulations, the mean fuel flow rate is maintained constant by adjusting the opening of a needle valve and the forcing frequency is varied from 10 Hz to 1 kHz in about 10 Hz increments. During these measurements the variations in the mean fuel pressure and the CH^* chemiluminescence intensity are found to be within 1.5%. The sampling frequency is 5 kHz, and the data length is 8 s. The data are decomposed into eight segments, and fast Fourier transform (FFT) analysis is performed for each second. Without explicit denotation, FTFs are the average of the results for the eight data segments.

Figure 2 shows the amplitude and phase of the measured flame transfer function $W_0(s)$ of up to 1 kHz. The FTF amplitude consistently increases up to 600 Hz, peaks around 750 Hz, and then drops. However, different trends are observed between 600 Hz and 660 Hz. This is caused by relatively strong acoustic responses to fuel modulations within this frequency range. The blocking ratio for the inlet and exit of the combustion chamber is 88% and 95%, respectively. Thus the combustion chamber can be considered as acoustically closed at both ends. Note that the blocking ratio is defined as the ratio of the solid surface area that air cannot flow through to the total sectional area normal to the air-flowing direction. The blocking ratio is zero downstream of the quartz tube except at the exit of the combustion chamber where a choking plate is installed. The one-wave-mode acoustic frequency of the combustion chamber is around 630 Hz. Without the baffle plates, self-excited combustion instability occurs around 630 Hz, with an amplitude of up to 12 kPa. Although the baffle plates significantly enhance acoustic damping, relatively large-amplitude

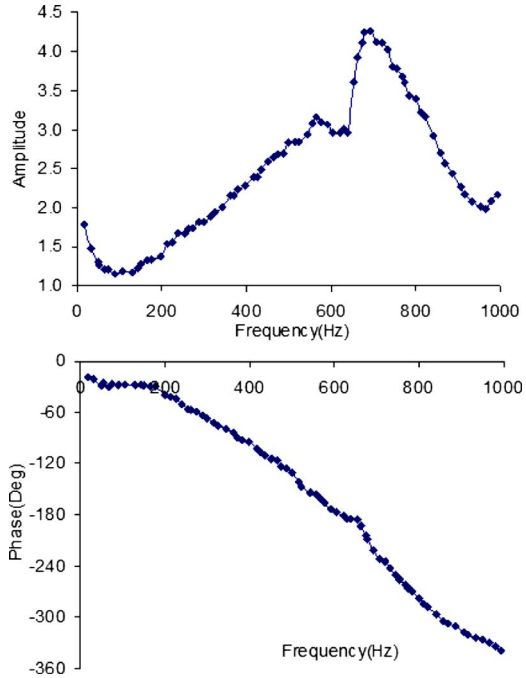

Fig. 2 Flame transfer function $W_0(s)$. The mean fuel pressure is 453 kPa, the air flow rate is 66.7 g/s, the equivalence ratio is 0.34, and the preheat temperature is 473 K. The fuel-modulation ratio is typically below 2.0%. The combustor is fueled with 1-Decene.

acoustic oscillations still occur with fuel modulations around this resonant frequency. This, in fact, implies that the measured FTFs between 600 Hz and 660 Hz are no longer open-loop transfer functions. The present paper's focus is on open-loop transfer functions below 600 Hz. Within this frequency range, forcing-induced acoustic oscillation intensities are small. For these experiments, the ratio of the complex amplitude $\tilde{A}/\tilde{B}$ is determined as 1.80. The gain of W_N consistently decreases with frequency of up to 0.65 at 600 Hz, while the phase lead consistently increases with frequency of up to 20.5 deg at 600 Hz.

4.1 Signal-to-Noise Ratio. The measured fuel pressure and CH^* chemiluminescence intensity are inevitably contaminated by electrical noises, external disturbances, and quantization errors. CH^* chemiluminescence intensity is intrinsically of broadband or stochastic nature, because of the wide range of spatial and temporal scales in flame/turbulence interactions. For reliable FTFs to be developed, fuel-modulation-induced chemiluminescence oscillations must be stronger than the background noise, i.e., the signal-to-noise ratio (SNR) must be sufficiently high. In our experiments, a SNR above 20 dB can be achieved with a fuel-modulation ratio larger than 0.5%.

4.2 Repeatability and Nonlinearity. The measured FTF is repeatable not only within the same sequence of experiments (limited by the requirements on paper length, results are not shown) but also among experiments conducted at different days (see Fig. 3). Even in the presence of self-excited large-amplitude combustion oscillations (up to 12 kPa in amplitude) around 640 Hz, the FTF at frequencies below 600 Hz is almost the same as those shown in Fig. 3. Strong repeatability in the FTF implies that a meaningful and reliable FTF has been obtained. One can see that below 600 Hz, the gain of the FTF for Jet-A is almost the same as that for 1-Decene, while above 320 Hz the phase difference slightly increases with frequency of up to 15 deg at 600 Hz. Since the droplets evaporation is not the limiting process, differences in FFTs between Jet-A and 1-Decene are associated with chemical kinetics. Jet-A has a considerable amount of hydrocarbons, which

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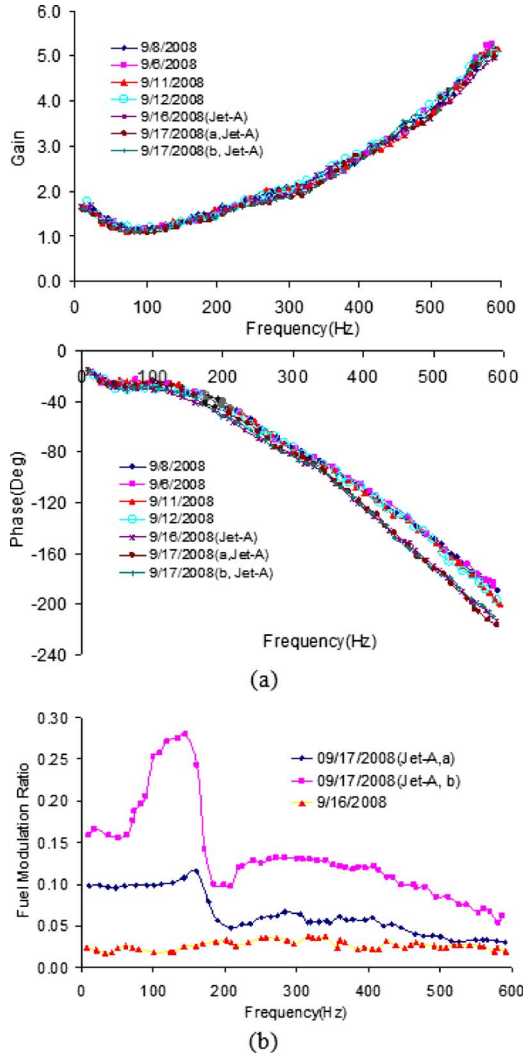

Fig. 3 (*a*) Flame transfer function $W_m(s)$ measured at different days and (*b*) the fuel-modulation ratio. The mean fuel pressure is 453 kPa, the air flow rate is 66.7 g/s, the equivalence ratio is 0.34, and the preheat temperature is 473 K. The fuel-modulation ratio is typically below 2.0%.

are longer than 1-Decene. A larger hydrocarbon molecule has a larger molecular weight, so its diffusion toward the reaction zone is smaller, and the chain-initiation will require more energy. Both slow down the chemical reaction rates and the heat release responses to fuel modulations.

Possible nonlinearity in the forced flame responses is checked by examining FTFs measured at different fuel-modulation amplitudes. It is believed that the saturation thresholds, above which nonlinearity starts to appear, depend on the forcing frequency and working conditions, such as the equivalence ratio, the preheat temperature, and the air flow rates. It is found that linearity in the forced flame response is well kept, since the measured FTFs are not sensitive to the fuel-modulation ratio (see Fig. 3(*b*)).

4.3 Low-Order Modeling. The gain and phases of the measured FTF shown in Fig. 3 can be modeled by a low-order rational transfer function. Such low-order models are very convenient for combustion analysis and control. However, before the low-order models are developed, it is worthwhile to further explore system dynamics from the measured FTF.

A plateau appears in phases of the FTF between 50 Hz and 200 Hz. This is mainly because within this frequency range, forcing cycles are much longer than the time scales of fuel convection,

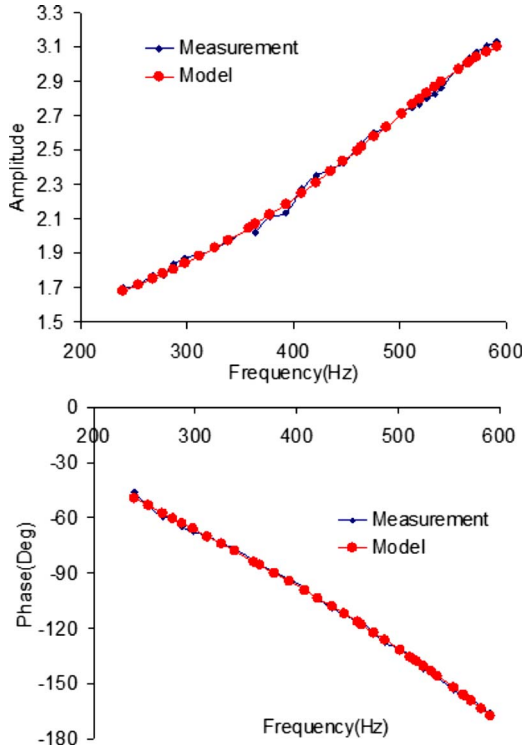

Fig. 4 Flame transfer function $W_0(s)$ versus Bode plots of the low-order model

evaporation, and chemical kinetics, as shown in Ref. [9]. The flatness in phases below 200 Hz implies the smallness of time delay in the forced flame responses. Phase lag associated with a pure time-delay unit is proportional to frequency, and the gain is unity. Since the time delay is small, the phase lag should be associated with nontime-delay units, such as a rational transfer function. Rational transfer functions allow both the gain and phases to vary continuously with frequency, which are very convenient for stability analysis and control. A low-order model is developed from the measured $W_0(s)$ from 240 Hz to 600 Hz,

$$W(z^{-1}) = \frac{\mathrm{CH}^*(z^{-1})}{P_0(z^{-1})} = \frac{-0.4213z^{-2} + 1.604z^{-3} - 1.147z^{-4}}{1 - 1.952z^{-1} + 1.525z^{-2} - 0.5222z^{-3}} \quad (5)$$

As shown in Fig. 4, the low-order model (5) closely matches the measured FTF in both amplitude and phases. The differences are less than 0.5% in amplitude and within 0.5 deg in phases. The FTF measured on September 12, 2008 are used for model development here. The measured FTF is decomposed into two data segments. One segment is used for model development, and the other one is used for validation. The time delay is identified as only one tap, i.e., 0.2 ms. The sampling frequency is 5 kHz. Small time delay greatly facilitates combustion analysis and control.

4.4 FTFs at Different Air Flow Rates (the Same Fuel Flow Rate). Figure 5 shows the flame transfer function $W_0(s)$ at three air flow rates, 58.4 g/s (ϕ=0.39), 66.7 g/s (ϕ=0.34), and 75.1 g/s (ϕ=0.30), respectively. The fuel flow rate is the same, 1.4 g/s. The air flow rate affects the FTF in two ways. First, it modifies the velocity field, the flame structure, droplets dispersion, fuel/air mixing, and the residence time (τ) and second, at the same fuel flow rate, the equivalence ratio is lower at a higher air flow rate, which slows down chemical kinetics, fuel evaporation, and so on. The increase in phase lag at lower flame temperature can be explained by larger time scales of evaporation (τ_V) and chemical kinetics (τ_c). At an air flow rate of 75.1 g/s, there is no plateau in phases below 200 Hz, which is different from that at 66.7 g/s and 58.4 g/s. This is because, at an air flow rate of 75.1 g/s (ϕ

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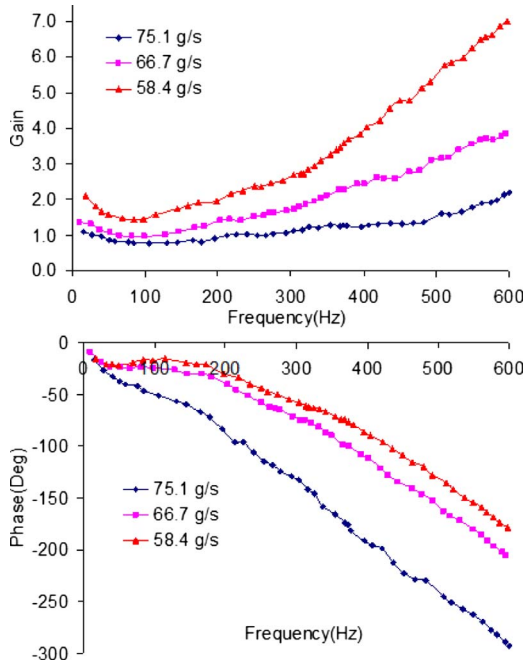

Fig. 5 Flame transfer function $W_0(s)$ at different air flow rates. The fuel-modulation ratio is typically below 2.0%, the preheat temperature is 473 K, and the fuel flow rate is 1.4 g/s.

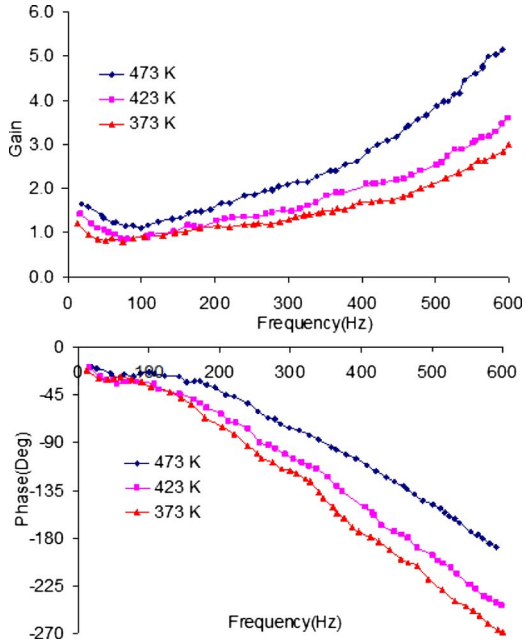

Fig. 6 Flame transfer function $W_m(s)$ at different preheat temperature. The air flow rate is 66.7 g/s, and the equivalence ratio is 0.31.

=0.30), the time scales of evaporation and chemical kinetics have considerably increased, and become comparable with the forcing cycle. Different from the phases, root-cause analysis of the gain decrease at lower flame temperature is less straightforward. For spray flames, the Damköhler number D_a can be defined as $D_a = \tau/(\tau_c + \tau_V)$. The residence time decreases almost linearly with the air flow rate, but the time scales of evaporation and chemical kinetics increase almost exponentially. With a small D_a or a large strain rate, the spray may not have enough time to evaporate, and chemical kinetics is slower at lower flame temperature. This is equivalent to a larger inertia in system dynamics. At lower flame temperature, the flame just cannot respond to fuel modulations as fast as at higher flame temperature, which inevitably results in smaller gain in the FTF. In the present experiments, the frequency responses are linear in terms of the fuel-modulation ratio but nonlinear to the flame temperature. This can be seen from the increasing differences in the FTF above 350 Hz among different air flow rates. The nonlinearity of the FTF to the flame temperature is not surprising, since both evaporation and chemical kinetics are highly nonlinear functions of the flame temperature.

4.5 FTFs at Different Preheat Temperature. The effects of preheat temperature and the equivalence ratio on forced flame responses are essentially similar, since they affect the flame temperature in the same direction. Note that at higher preheat temperature, the air velocity at the swirler exit is higher, which enhances droplet evaporation and fuel/air mixing. Figure 6 shows the flame transfer function $W_m(s)$ at the preheat temperature of 473 K, 423 K, and 373 K, respectively. The air flow rate is 66.7 g/s, and the equivalence ratio is 0.31. At the same air flow rate, the air velocity is larger at higher preheat temperature, thus the residence time decreases, but the time scales of evaporation and chemical kinetics decreases even further. Then the flame becomes more responsive to fuel modulations, resulting in larger gain and a smaller phase lag in the FTF. The FTF is nonlinear in terms of the preheat temperature, which is most obvious for fuel modulations above 400 Hz. From Figs. 5 and 6, one can see that the effects of the equivalence ratio and preheat temperatures on the forced flame responses are essentially similar.

5 Applications of FTFs

The FTF can be directly used for analysis and control of combustion instability and LBO. Active control of both phenomena can be achieved using small-amplitude fuel modulations, employing the same control hardware and fuel actuators. However, several major differences exist between combustion instability control and LBO control. First, combustion instability is linearly unstable. But near-LBO combustion dynamics is linearly stable. Stable combustion cannot be maintained at all if the equivalence ratio drops below certain thresholds. Second, because combustion instability is linearly unstable, the fuel actuator should keep working to prevent acoustic pressure from regaining energy, even after the initial strong pressure pulsations have been suppressed. However, for LBO control, no further fuel actuations are required after the initial deviations from the equilibrium points have been diminished. Third, combustion instability is temporally and spatially coherent and occurs around the acoustic resonant frequency. But near-LBO combustion dynamics are typically much less coherent and usually occur below 200 Hz. This implies that for LBO control, the actuator bandwidth and time delay associated with fuel convection, evaporation, and chemical kinetics may not be major technical challenges.

Stability enhancement of near-LBO combustion using a simple linear-quadratic-Gaussian (LQG) feedback controller has been numerically demonstrated in Ref. [4]. Such feedback controllers can also be designed based on the measured FTFs. Since an aero-engine has to operate within a large range of working conditions, stable adaptive controllers, which have superior robust stability and control performance, are promising. The measured near-LBO flame transfer functions have no unstable zeros, no unstable poles, and even negligibly small time delay, thus adaptive robust controller can be easily worked. Besides the liquid-fueled lean direct fuel injection (LDI) combustors, the developed control strategy can also be used for gas-fueled combustion, such as syngas- and natural-gas-fueled combustion. Because of variations in feedstocks, the ratio of H_2/CO in syngas can change from 0.33 to 40, and the concentration of water vapor can vary from 0% to 40% [12]. This will impose significant challenges to stable combustion in particular at very lean conditions. Near-LBO combustion dynamics may exhibit persistent low-frequency oscillations [13,14]

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or undergo partial or global quenching without showing much oscillatory behavior. Both situations can be handled by the proposed feedback controllers.

6 Conclusions

A method for determining the instantaneous fuel flow rate out of a liquid fuel nozzle is developed, which is based on acoustic wave theory and pressure measurements upstream of the fuel nozzle. Repeatability and linearity in the measured FTF are demonstrated. The derived FTF is not sensitive to variations in the fuel-modulation amplitude. Low-order models in the form of a rational transfer function are developed from the measured gain and phases, which can be conveniently used for combustion analysis and control. Effects of the equivalence ratio and preheat temperature on the FTF are examined. The phase lag decreases with flame temperature and the gain increases. Heat release rate responses are linear in terms of the fuel-modulation amplitude, but nonlinear in terms of the flame temperature. Applications of the FTF to combustion control, including combustion instability and lean blowout, are discussed.

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Nomenclature

$\tilde{A}$ and $\tilde{B}$ = complex amplitude of acoustic waves
$\bar{c}$ = mean speed of sound, m/s
D_a = the Damköhler number
i = complex symbol
$k=\omega/\bar{c}$ = wave number
$\dot{m}'(t)$ = fuel modulations, kg/s
$\bar{\dot{m}}_f$ = the mean fuel flow rate out of a fuel injector, kg/s
$\tilde{P}(0,t)$ = the estimated complex dynamic pressure immediately upstream of the fuel nozzle, Pa
$p_0(t)$ = the estimated dynamic pressure immediately upstream of the fuel injector, Pa
$p_m(t)$ = dynamic pressure measured at 0.29 m upstream of the fuel injector, Pa
$s=i\omega$ = symbol of the Laplace transform
$W_N(s)$ = the transfer function between pressure measurements and pressure immediately upstream of a fuel nozzle
x_m = a pressure measurement location, 0.29 m upstream of the fuel nozzle, m
z = symbol of Z transform
ϕ = equivalence ratio
$\mathrm{Re}(\ldots)$ = the real part of a complex quantity
$\overline{\Delta P}$ = mean pressure drop across a fuel nozzle, Pa
ω = angular frequency, radians/s
τ = the residence time, s
τ_c = the time scale of chemical kinetics, s
τ_V = the evaporation time of droplets, s

Abbreviation

ICCD = intensified charge-coupled device

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